KOLKATA FREIGHT & LOGISTICS (KFL) – EXECUTIVE MANAGEMENT MEMORANDUM

ROOT-CAUSE ANALYSIS & PERFORMANCE INTELLIGENCE INVESTIGATION | FY 2024

TO: Board of Directors & Chief Operating Officer
DATE: November 6, 2024
FROM: Senior Analytics & Performance Intelligence Team
SUBJECT: Headline OTD Decline Root-Cause Attribution

1. Executive Summary & Business Problem

During the second half of FY 2024 (Q3–Q4), Kolkata Freight & Logistics experienced a severe collapse in headline operational performance. The primary business metric, **On-Time Delivery Rate (OTD)**, deteriorated from an H1 baseline of **90.26%** (exceeding the corporate 85.0% SLA commitment) to **64.30%** in H2, representing an absolute drop of **-25.96 percentage points** (a -28.8% relative contraction). This operational failure resulted in elevated contract SLA penalties, a surge in enterprise client churn, and severe warehouse sorting congestion.

2. Empirical Driver Tree Decomposition (Target: >= 80% Explained)

Using counterfactual standardization and marginal factor attribution across 349,553 completed consignments and 1.14 million scan events, our model successfully accounts for **92.7% of the observed decline** without double-counting:

Operational Driver	Mechanism	Decline Impact	Share (%)
Warehouse Sorting Bottlenecks	Conveyor breakdowns & sorting backlogs at Dhulagarh (DHU) & Taratala (TRT)	**-8.94 pp**	34.4%
Corridor Transit Delays	Bridge repairs & construction congestion on NH16, Kona Exp, and Dankuni bypass	**-7.62 pp**	29.4%
Vehicle Breakdowns & Downtime	Deferred preventative maintenance in 14ft LCV & 32ft multi-axle aging fleet	**-4.38 pp**	16.9%
Service Mix Shift	Expansion of tight-window Express/Same-Day services (8% to 16% volume share)	**-1.85 pp**	7.1%
Festive Surge Shock (W39-44)	Temporary Durga Puja freight spike (+45% volume) causing transient capacity strain	**-1.28 pp**	4.9%
Unexplained Residual	Stochastic transit noise, localized loading dock variances, and adverse weather	**-1.89 pp**	7.3%
Total Net Headline Decline	**Combined cumulative impact across network**	**-25.96 pp**	**100.0%**

3. Deep-Dive Findings on Primary Operational Bottlenecks

- **Dhulagarh & Taratala Hub Failure (Hypothesis H4 Supported):** Facility performance was highly asymmetric. While Salt Lake (HUB_SLK) and Kalyani (HUB_KLY) maintained >82% OTD, Dhulagarh collapsed by 33.2 pp and Taratala by 29.8 pp. Origin sort dwell time at these facilities increased from 36.4 minutes to 118.5 minutes ($t=48.2$, $p<0.001$).
- **Highway Infrastructure Transit Friction (Hypothesis H1 Supported):** Arterial routes passing through the Kona Expressway corridor averaged 2.45 hours of transit delay versus 0.65 hours on peripheral routes (Welch $t=31.4$, $p<0.001$), turning planned 2-hour hauls into 5+ hour trips.
- **Aging Haulage Fleet Mechanical Risk (Hypothesis H2 Supported):** Logistic regression demonstrates that vehicles with >40 hours of downtime increased consignment delay probability by 3.4x (Odds Ratio = 3.42, 95% CI [3.12, 3.75]). Over 72% of breakdowns were concentrated in an identifiable 20% aging vehicle cohort.

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4. Analysis of Refuted Hypotheses: The Seasonality Myth

Hypothesis H6 (Seasonal Demand as Root Cause) is REJECTED: Management initially suspected that the annual Durga Puja and Diwali festival surge was the primary culprit behind missed delivery SLAs. However, statistical phase decomposition reveals that OTD had **already dropped to 65.4% during Weeks 27–38**, well before the festive volume surge began in Week 39. Furthermore, when shipment volume returned to normal baseline levels in Weeks 45–52, OTD failed to recover (remaining at 63.8%). While festival volume added acute short-term strain, it was NOT the structural cause of the multi-quarter performance collapse.

5. Targeted Operational Action Matrix

Focus Area	Operational Intervention	Responsible Owner	Measurable Target	Timeline	Review Cadence
Warehouse Automation	Overhaul sorting lines at Dhulagarh & Taratala; deploy secondary manual sorting bypass	VP Warehouse Operations	Sort dwell time < 45 min; zero backlog	6 Weeks	Weekly
Dynamic Highway Routing	Implement GPS geofenced routing bypassing Kona Expressway during 08:00-12:00 peak	Head of Fleet Dispatch	Arterial delay < 40 min per trip	3 Weeks	Bi-Weekly
Fleet Preventative Care	Mandatory 10,000 km overhaul on aging multi-axle fleet; retire 50 highest-downtime LCVs	Fleet Maintenance Director	Fleet breakdown rate < 1.0%	8 Weeks	Monthly
Express SLA Pricing	Re-calibrate Same Day delivery windows from 6h to 8h in congested zones or apply congestion tariff	Chief Commercial Officer	Same Day OTD >= 85.0%	4 Weeks	Monthly
Early-Warning Deployment	Operationalize automated weekly composite risk scoring index with 4-week lead alert	Head of Data & BI	Zero undetected bottleneck triggers	2 Weeks	Weekly

6. Early-Warning Intelligence & Leading Indicators

Backtesting our multi-factor Early-Warning System over 52 historical weeks demonstrated that leading indicators (warehouse sort duration creep and arterial transit delays) triggered an **Orange Warning at Week 23**—a full **4 weeks prior to the public collapse of OTD in Week 27**. By tracking composite operational risk rather than lagging OTD, management can execute localized load-shedding and fleet reassignments before contractual SLAs are breached.

7. Methodological Limitations & Synthetic Data Disclosure

DATA NOTICE: *This memorandum is part of a portfolio demonstration based on a fictional logistics network in Kolkata. All 1,546,782 records across shipments, routes, vehicles, and scans were synthetically generated with fixed random seeds for reproducibility. Operational findings represent modeled scenarios and must never be cited as real-world company records. Analytical limitations include lack of driver-level telematics and micro-weather precipitation telemetry.*